

ECE3300L Variable PWM Project

Reusable Edge-Aligned PWM Generator: 100 MHz Clock, 25 kHz Output

Project direction

In this assignment, you will design the specification for a reusable variable-duty PWM core. The design is divided into a period timebase and a duty comparator so later projects can provide commands from switches, sensors, processors, or control algorithms without changing the PWM timing logic.

What you will build

1. A parameterized PWM timebase that divides the 100 MHz system clock into fixed 4000-clock periods.
2. A period counter that exposes the current phase from 0 through 3999 and identifies the start of each new period.
3. A variable-duty comparison module whose input selects how many clocks the output remains high.
4. A boundary-update mechanism that prevents a changing command from creating a shortened or extended pulse in the active period.
5. A reusable, active-high PWM core with exact behavior at 0% and 100% duty cycle.

PWM configuration for this assignment

Setting	Required value	Meaning
System clock	100 MHz	All registers remain synchronous to the FPGA clock.
PWM frequency	25 kHz	One output period occurs every 40 microseconds.
Period length	4000 clocks	The phase counter represents values 0 through 3999.
PWM alignment	Edge-aligned	Every high pulse begins at the start of a period.
Duty command	0 through 4000	The command represents the requested number of high clocks.
Output polarity	Active high	The asserted portion of the PWM period is logic 1.

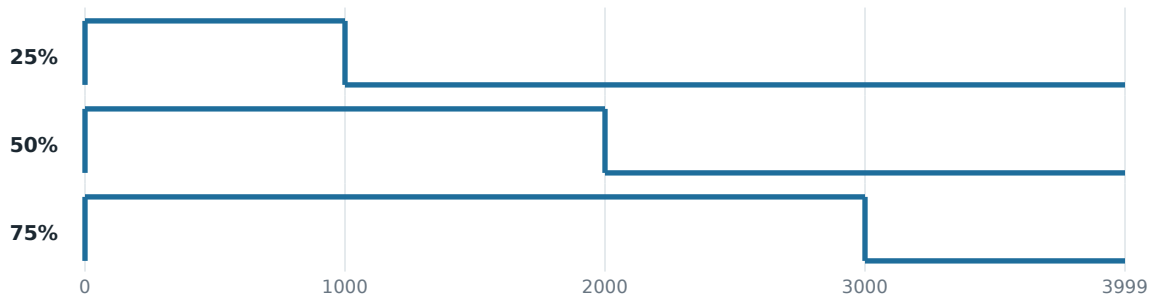
Lab objectives

- Distinguish PWM frequency from PWM duty cycle.
- Translate a desired output frequency into an exact system-clock counter range.
- Generate variable pulse widths while preserving a constant output period.
- Separate shared timing logic from duty-cycle comparison logic.
- Update a control value at a safe boundary to prevent malformed pulses.

PWM timing and duty cycle

Pulse-width modulation represents a control value through the fraction of each fixed period spent high. Changing duty cycle changes pulse width; it does not change the PWM frequency.

Edge-aligned PWM: constant period, variable high time



Frequency and duty calculations

Period calculation

For a 100 MHz clock and 25 kHz PWM output: $100,000,000 / 25,000 = 4000$. Therefore, the timebase counts exactly 4000 clocks per period, and each period lasts **40 microseconds**.

Requested duty	High-clock count	Output behavior
0%	0	Always low; the comparison is never true.
25%	1000	High for counts 0 through 999.
50%	2000	High for counts 0 through 1999.
75%	3000	High for counts 0 through 2999.
100%	4000	Always high because every valid phase count is below 4000.

Core comparison rule

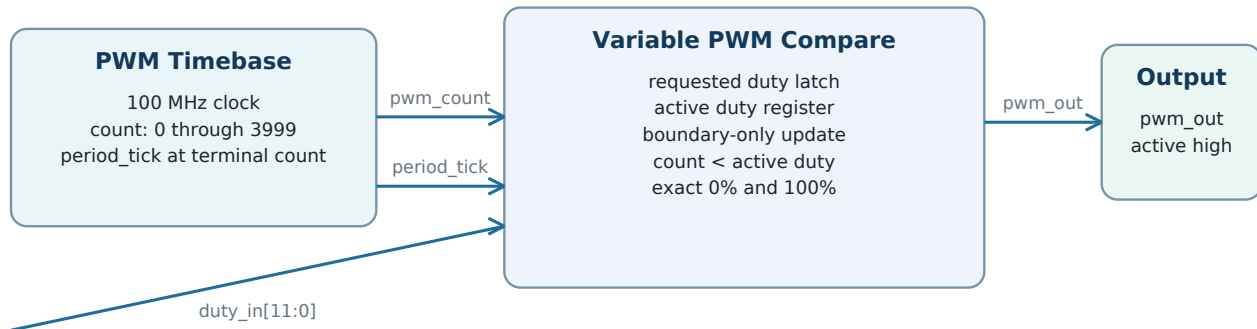
During a period, drive `pwm_out` high when `pwm_count < active_duty`; otherwise drive it low. Because the timebase never reaches 4000, this single rule naturally supports both endpoint duty cycles.

Core module interfaces

Module	Signal	Direction	Purpose
pwm_timebase	clk	Input	100 MHz system clock.
pwm_timebase	rst_n	Input	Active-low reset for the phase counter.
pwm_timebase	pwm_count[11:0]	Output	Current phase position from 0 through 3999.
pwm_timebase	period_tick	Output	High at count 3999; the next clock edge begins a new PWM period.
variable_pwm	clk, rst_n	Inputs	Shared clock and active-low reset.
variable_pwm	pwm_count[11:0]	Input	Phase count supplied by the timebase.
variable_pwm	period_tick	Input	Permits the requested duty value to become active at the wrap edge.
variable_pwm	duty_in[11:0]	Input	Requested high time, expressed in clocks from 0 through 4000.
variable_pwm	pwm_out	Output	Active-high edge-aligned PWM waveform.

Module-level architecture

The two-module structure keeps frequency generation independent from the source and meaning of the duty command.



Reset and the 100 MHz system clock are shared by both modules.

A later top module may translate switches, a sensor controller, or another command source into **duty_in**.

Timebase behavior

1. After reset, **pwm_count** begins at zero.
2. On each 100 MHz clock edge, the count advances by one while remaining within the range 0 through 3999.
3. After count 3999, the next value is zero. This wrap begins the next 40-microsecond PWM period.
4. Assert **period_tick** while the counter is at 3999. On the next clock edge, the counter wraps to zero and a new period begins.

Variable-duty behavior

1. Treat **duty_in** as a requested value. It may change while the present PWM period is active.
2. When **period_tick** is asserted, copy the requested value into an internal active-duty register on the same edge that the phase counter wraps. The active value then remains unchanged for the entire new period.
3. If a requested value is greater than 4000, saturate the active value to 4000 rather than allowing an undefined command range.
4. Compare **pwm_count** with the active-duty register throughout the period. The output begins high at count zero and returns low when the count reaches the active threshold.
5. A new requested duty takes effect only at the next period boundary, so every completed pulse belongs entirely to one command value.

Design requirements and boundaries

The completed core must remain predictable at its endpoint values, reusable at other parameter settings, and independent of any application-specific load.

Required design rules

- Use the 100 MHz input as the only clock. The PWM output and period_tick are signals, not additional clocks.
- Derive period length from clock-frequency and PWM-frequency parameters rather than scattering 4000 through the design.
- Size the phase count for 0 through 3999 and the duty representation for 0 through 4000.
- Reset the phase counter, active-duty register, and PWM output to known values.
- Keep PWM frequency constant when the duty request changes.
- Preserve exact endpoint behavior: duty 0 must remain low and duty 4000 must remain high.
- Use one consistent active-high polarity inside the core. Any external inversion belongs outside these modules.

Parameter relationships

Quantity	Relationship	Assignment value
Period length	system clock / PWM frequency	4000 clocks
Maximum phase	period length - 1	3999
Maximum duty	period length	4000
Phase-counter width	Enough bits for 0 through maximum phase	12 bits
Duty-register width	Enough bits for 0 through maximum duty	12 bits

Command-change examples

Event	Current active duty	Requested duty	Required result
Request changes halfway through a period	25%	75%	Finish the current period at 25%; begin 75% at the next boundary.
Request is stable before count 3999	50%	25%	Capture 25% on the wrap edge and keep it active from count 0 through 3999.
Request is above the valid range	Any	Greater than 4000	Capture 4000 so the next period is continuously high.
Request becomes zero	Any	0	At the next boundary, produce a continuously low period.

Design scope

Core-only assignment

Create only the reusable PWM timebase and variable-duty generator specification. A board-level top module, switch mapping, displays, sensors, motor or power drivers, closed-loop control, and application-specific behavior are intentionally outside this assignment.

End of assignment

The completed design scope is a parameterized, edge-aligned PWM core whose frequency remains fixed while its high time changes cleanly at period boundaries.